Course Outline

Sem		Course Name	Th	Tut	Pra	Hrs	Cr
Sem I	HSC	Humanities – 1	2	0	0	2	2
	BSC	Mathematics – 1	3	1	0	4	4
	BSC	Physics	3	0	0	3	3
	BSC	Physics – Laboratory	0	0	2	2	1
	BSC	Environment Science	2	0	0	2	2
	ESC	Workshop Practices	0	0	2	2	1
	BSC	Biology for Engineers	2	0	0	2	2
	ESC	Computer Programming -I	1	0	0	1	1
	ESC	Computer Programming -I Laboratory	0	0	2	2	1
	HSC	Universal Human Values	1	0	0	1	1
	HSC	Indian Knowledge System	2	0	0	2	2
			16	**1**	**5**	**23**	**20**
Sem II	HSC	Humanities – 2	2	0	0	2	2
	BSC	Mathematics – 2	3	1	0	4	4
	BSC	Chemistry	3	0	0	3	3
	BSC	Chemistry – Laboratory	0	0	2	2	1
	ESC	Introduction to Electrical Engineering	3	0	0	3	3
	ESC	Introduction to Electrical Engineering-Laboratory	0	0	2	2	1
	ESC	Engineering Graphics	0	0	4	4	2
	HSC	Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC	Organizational Behaviour	1	0	0	1	1
	ESC	Computer Programming -II	1	0	0	1	1
	ESC	Computer Programming -II Laboratory	0	0	2	2	1
			13	**1**	**12**	**26**	**20**
Sem III	Pro	Civic and Social Service Internship	0	0	0	0	1
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	PC	Digital Circuits	3	0	0	3	3
	PC	Digital Circuits Laboratory	0	0	2	2	1
	PC	Electronics Devices and Circuits	3	0	0	3	3
	PC	Electronics Devices and Circuits Laboratory	0	0	2	2	1
	PC	Networks and Systems	3	1	0	4	4
	BSC	Math-3	3	1	0	4	4
			15	**2**	**4**	**21**	**20**
Sem IV	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 – Laboratory (Department Specific)	0	0	2	2	1
	OE	Open Elective 1 (From Other School)	3	0	0	3	3
	PC	Analog Electronics	3	0	0	3	3
	PC	Analog Electronics Laboratory	0	0	2	2	1
	PC	Analog Communication	3	0	0	3	3
	PC	Analog Communication Laboratory	0	0	2	2	1
	PC	Electromagnetics and Transmission Lines	3	1	0	4	4
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3

Semester 4

<Course Code>					Analog Electronics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To introduce circuit realizations with bjts.
- To give understanding of various types of amplifier circuits such as small signal, cascaded, large signal and tuned amplifiers.
- To familiarize the concept of feedback in amplifiers so as to differentiate between negative and positive feedback.
- to evolve the students as oscillator designers.

UNIT I: BJT AMPLIFIER	**8 Hrs.**
Introduction to biasing of BJT, operating point, bias stability for different circuits, stabilization against V_{BE}, I_{CO} and β, bias compensation techniques, thermal runaway, thermal stability.	
UNIT II: SMALL SIGNAL MODEL	**14 Hrs.**
Two port device and Hybrid model, Analysis of CE, CC, and CB Amplifiers, CE Amplifier with emitter resistance, low frequency response of BJT Amplifiers, Miller‟s Theorem, Dual of Miller‟s theorem, Differential amplifier using BJT.	
UNIT III: MULTISTAGE AMPLIFIER	**10 Hrs.**
Multistage amplifier, coupling mechanism, effect of coupling and bypass capacitors, Design of single stage and multistage RC coupled amplifier, Analysis of Cascaded RC Coupled amplifiers, Cascode amplifier, Darlington pair, Current mirrors, High frequency pi Model, High frequency analysis.	
UNIT IV: FEEDBACK AMPLIFIER AND OSCILLATORS	**10 Hrs.**
Introduction, types of amplifier, concept of feedback, feedback topology, effect on gain, bandwidth, input and output resistance, Use of positive feedback, Barkhausen criterion for oscillations, sinusoidal oscillators, phase shift, resonant circuit oscillators, crystal oscillator. Different oscillator circuits-tuned collector, Hartley Colpitts, phase shift, Wien‟s bridge, and crystal oscillator.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember the concepts bjt related design issues.
CO2	:	Understand the analog circuit and its design.
CO3	:	Apply small signal amplifier model for different circuits.
CO4	:	Analyse cascaded amplifier configurations to obtain the required overall specifications.
CO5	:	Design different amplifiers.
CO6	:	Build analog subsystem using amplifier and oscillator.

TEXT/REFERENCE BOOKS

1. Integrated electronics, jacob millman, christos c halkias, mcgraw hill education
2. Electronic devices and circuits theory– robert l. Boylestead, louis nashelsky, 11th edition, 2009, pearson.
3. Electronic devices conventional and current version -thomas l. Floyd 2015, person
4. Electronic devices and circuits, david a. Bell – 5th edition, oxford.
5. Electronic devices and circuits, s. Salivahanan, n. Suresh kumar, a vallvaraj, 5th edition, mc graw hill education.
6. Electronics circuits and applications, md. H rashid, cengage, 2014.

<Course Code>					Analog Electronics Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To experiment with circuit realizations with bjts.
- To give understanding of various types of amplifier circuits such as small signal, cascaded, large signal and tuned
- Amplifiers.
- To familiarize the concept of feedback in amplifiers so as to differentiate between negative and positive feedback.
- To design different oscillator

LIST OF EXPERIMENTS	
1	To study components of ae-1 lab.
2	To study common emitter amplifier.
3	To study different biasing circuits of bjt.
4	To study transfer and drain characteristic of fet and mosfet.
5	To obtain operating point of the given circuit.
6	To study thermal stability of the given circuit.
7	To measure the hybrid parameter of bjt.
8	To obtain frequency response of single stage ce amplifier.
9	To study the effect of negative feedback with and without bypass capacitor.
10	To obtain frequency response of two stage rc coupled amplifier.
11	To study feedback amplifier.
12	To study rc phase shift oscillator.
13	To study resonant oscillator.
14	To design of mini project in a group of 4-5 students.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember the concepts bjt related design issues.
CO2	:	Understand the analog circuit and its design.
CO3	:	Apply small signal amplifier model for different circuits.
CO4	:	Analyse cascaded amplifier configurations to obtain the required overall specifications.
CO5	:	Design different amplifiers.
CO6	:	Build analog subsystem using amplifier and oscillator.

TEXT/REFERENCE BOOKS

1. Integrated electronics, jacob millman, christos c halkias, mcgraw hill education
2. Electronic devices and circuits theory– robert l. Boylestead, louis nashelsky, 11th edition, 2009, pearson.
3. Electronic devices conventional and current version -thomas l. Floyd 2015, person
4. Electronic devices and circuits, david a. Bell – 5th edition, oxford.
5. Electronic devices and circuits, s. Salivahanan, n. Suresh kumar, a vallvaraj, 5th edition, mc graw hill education.
6. Electronics circuits and applications, md h rashid, cengage 2014.

24ECE***T					Analog Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To learn the fundamentals of a communication system.
- To be able to understand, analyze, and design analog communication systems
- To implement analog transmitter-receiver modules and analyze its performance.
- To study different analog communication systems in the presence of noise in channel.

UNIT I: RANDOM VARIABLE AND RANDOM PROCESS	**10 Hrs.**
Classical and axiomatic definitions of probability, probability of an event, addition rule and conditional probability, multiplication rule, Bayes‟ theorem and independence, Discrete and continuous random variable: definitions and examples, Probability density function and cumulative distribution functions of continuous random variables, Probability mass function of discrete random variables, expected values and variance of discrete random variable, Statistical averages: Function of Random variables, Moments, Mean, Correlation and Covariance function: Principles of autocorrelation function, cross – correlation functions. Central limit theorem, Properties of Gaussian process. Measure of central tendency:	
UNIT II: AMPLITUDE MODULATION	**14 Hrs.**
Introduction: General architecture of a communication system, AM: Time-Domain description, Frequency – Domain description. Generation of AM wave: square law modulator, switching modulator. Detection of AM waves: square law detector, envelop detector. Double side band suppressed carrier modulation (DSBSC): Time-Domain description, Frequency-Domain representation, Generation of DSBSC waves: balanced modulator, ring modulator. Coherent detection of DSBSC modulated waves, Superheterodyne Receiver, Costas loop. Quadrature carrier multiplexing, Hilbert transform, Pre-envelope, Canonical representation of band pass signals, Single Side Band (SSB) and Vestigial Side Band (VSB): Generation and Detection, comparison of amplitude modulation techniques.	
UNIT III: ANGLE MODULATION	**10 Hrs.**
Basic definitions, Phase Modulation and Frequency Modulation (PM and FM), Narrow band FM, Wide band FM, spectrum analysis of NBFM and WBFM, transmission bandwidth of FM waves, generation of FM waves: indirect FM and direct FM. Demodulation of FM waves, FM stereo multiplexing, Phase-locked loop, Nonlinear effects in FM systems	
UNIT IV: NOISE IN ANALOG COMMUNICATION SYSTEMS	**8 Hrs.**
Introduction, shot noise, thermal noise, white noise, Noise equivalent bandwidth, Noise Figure, Equivalent noise temperature. Noise in DSB-SC receivers, Noise in SSB receivers, Noise in AM receivers, Noise in FM receivers, Pre-emphasis and De-emphasis in FM, Auto correlation, Cross relation , Energy spectral density, Power spectral density and their applications in communication signal/noise analysis.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the basics of communication systems, analog transmitter and receiver design issues.
CO2	:	Analyze the functionality of various blocks of an analog communication system.
CO3	:	Design analog transmitter and receiver modules both at the block and circuit levels.
CO4	:	Evaluate various design issues and possible solution methods of analog communication systems.
CO5	:	Implement analog communication systems using BJT, FET and other electronic modules.
CO6	:	Build analog communication circuits, find design related issues, and measure its performance.

TEXT/REFERENCE BOOKS

1. Simon Haykins & Michael Moher, “Communication Systems”, 5th Edition, John Willey, India Pvt. Ltd, 2010
2. B. P. Lathi, “Modern Digital and Analog Communication Systems”, Oxford University Press., 4th edition.
3. Simon Haykins & Michael Moher, “An Introduction to Analog and Digital Communication”, John Wiley India Pvt. Ltd., 2008.
4. H.Taub & D.L.Schilling, “Principles of Communication Systems”, TMH, 2011.

24ECE***P					Analog Communication Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To learn the fundamentals of analog communication systems and its application for voice communications.
- To be able to analyze and design analog communication circuits.
- To be able to implement analog communication blocks using simulator (such as MATLAB) and analog circuit design simulators such as MultiSim and TINA.

LIST OF EXPERIMENTS	
1	Generation of Amplitude Modulated (AM) Signals for different values of modulation index
2	Demodulation of AM signals
3	Generation of DSBSC signal
4	Demodulation of DSBSC signal
5	Generation of SSBSC signal
6	Demodulation of SSBSC signal (Phase-shift method)
7	Generation of FM signal and finding its bandwidth using spectrum analysis
8	FM demodulation
9	Study the operation of Superheterodyne receiver
10	Study the operation of Phase-locked-loop (PLL)
11	Design and implementation of Pre Emphasis - De Emphasis Circuits
12	Design and implementation of Analog mixer

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember analog communication circuit design methodologies.
CO2	:	Understand the design methodologies to utilize them practically.
CO3	:	Apply design methods to realize analog communication circuits practically.
CO4	:	Analyze functionalities of the basic building blocks of analog communication system.
CO5	:	Evaluate and report performance of different analog communication techniques.
CO6	:	Design modulator and demodulator circuits using software.

TEXT/REFERENCE BOOKS

1. Simon Haykins & Michael Moher, "Communication Systems", 5th Edition, John Willey, India Pvt. Ltd, 2010
2. B. P. Lathi, "Modern Digital and Analog Communication Systems", Oxford University Press., 4th edition.
3. Simon Haykins & Michael Moher, "An Introduction to Analog and Digital Communication", John Wiley India Pvt. Ltd., 2008.
4. H.Taub & D.L.Schilling, "Principles of Communication Systems", TMH, 2011.

20ECE201T					Electromagnetics and Transmission Lines					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	0	0	100

COURSE OBJECTIVES

- To develop the fundamental understanding about electromagnetic waves.
- To provide the knowledge about concepts related to electrostatics and magneto statics.
- To understand electromagnetic wave characteristics.
- To provide knowledge of transmission line theory.

UNIT 1 Electrostatics	**10 Hrs**
Co-ordinate systems, Coulomb,,s law, Electric field Intensity, Fields due to different charge distributions, Electric Flux Density, Gauss law and its Applications, Electric Flux Density, Gauss law and its Applications, Electric Potential, Relation Between E and V, Maxwell,,s Two equations for Electrostatic Fields, energy Density, Maxwell,,s Two equations for Electrostatic Fields, energy Density, Illustrative Problems. Convection and Conduction Currents, Dielectric Constant, Isotropic and Homogeneous Dielectrics, Continuity Equation and Relaxation Time, Poisson,,s and Laplace,,s Equations, Capacitance- Parallel plate, Co-axial and Spherical capacitors, Illustrative Problems.	
UNIT 2: Magneto-statics	**10 Hrs.**
Biot-Savart Law, Ampere,,s circuital Law and Applications, Magnetic Flux Density, Maxwell‟s Two Equations for Magneto static fields, Magnetic Scalar and Vector Potentials, Forces due to Magnetic Fields, Ampere,,s force Law, Forces due to Magnetic Fields, Ampere,,s force Law, Forces due to Magnetic Fields, Ampere,,s force Law, Inductances and Magnetic Energy, Illustrative Problems. Maxwell,,s Equations (Time Varying Fields): Faraday,,s Law and Transformer emf, Inconsistence of Ampere,,s Law and Displacement Current density, Maxwell,,s Equations indifferent Final Forms and Word Statements, Conditions at a boundary Surface: Dielectric-dielectric, dielectric-conductor Interfaces, Illustrative Problems	
UNIT 3: EM Wave Characteristics	**11 Hrs.**
Wave Equations for conducting and Perfect Dielectric Media, Uniform Plane Waves-Definition, All Relations between E and H, Sinusoidal Variations, Wave Propagation in Lossless and Conducting Media, Conductors and Dielectrics-Characterization, Wave Propagation in good conductors and Good Dielectrics, Polarization, Illustrative Problems. Reflection and Refraction of Plane waves-Normal and Oblique Incidences for Perfect Dielectric, Brewster angle, Critical Angle, Total Internal Reflection, Surface Impedance, Poynting Vector Poynting Theorem-Applications, Power Loss in Plane Conductor, Illustrative Problems	
UNIT 4: Transmission Lines	**11 Hrs.**
Types, Parameters, Transmission line Equations, Primary and Secondary Constants, Expressions for Characteristic Impedance, Propagation Constant, Phase and Group Velocities, Infinite Line Concepts, Lossless ness/Low Loss Characterization, Distortion-Condition for Distortionlessness and Minimum Attenuation, Loading- Types of loading, Illustrative Problems. Input Impedance Relations, SC and OC Lines, Reflection Coefficient, VSWR, UHF Lines as Circuit Elements, λ/4, λ/2 and λ/8 Lines-Impedance Transformations, Significance of Zmin and Zmax, Smith Chart-Configuration and Applications, Single and Double Stub Matching, Illustrative Problems.	
	Max 42 Hrs.

COURSE OUTCOMES (CO):

CO1 - Remember different laws of electromagnetics
CO2 - Understand different concepts of electromagnetics and transmission lines
CO3 - Apply different concepts of electromagnetics and transmission lines to solve problems
CO4 - Analyze the different concepts electromagnetic and transmission line
CO5 - Evaluate the different electromagnetic and transmission line scenarios
CO6 – Create the application of electromagnetics and transmission lines

TEXT/REFERENCE BOOKS

1. Engineering electromagnetic-, William H. Hayt Jr. and John A. Buck, 7thEd. 2006, TMH.
2. Elements of electromagnetics, Matthew Sadiku, Oxford university press, 7th edition, 2018
3. Electromagnetic waves and radiating systems, E.C.Jordan, K.G.Balmain, 2nd edition, 2000, PHI
4. Electromagnetic Field Theory and Transmission Lines, G. S. N. Raju, Pearson Education India, 2006
5. Microwave Engineering, David M. Pozar, Wiley.

24ECE***T					Digital Signal Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To introduce mathematical ideas for analysis of discrete time signals and systems.
- To study various digital signal processing systems and its architecture.
- To understand methodology to analyze, design and implement various digital filters.
- To understand need and development of efficient algorithms for various DSP tasks.

UNIT I: DISCRETE TIME SIGNALS AND SYSTEMS	**11 Hrs.**
Continuous Time and Discrete Time Signals, Analog and Digital Signals – Definition and Examples, Classification of Discrete Time Signals, Energy, Power, Periodicity, Signal operations, Discrete time elementary signals, Discrete Time Systems - Definition, Classification, Discrete time LTI system and unit impulse (sample) response, Convolution sum, Linear constant co-efficient difference equation representation, Block Diagram Representation, Interconnections, Stability and causality, Sampling and Sampling theorem, Reconstruction, Aliasing, Concept of quantization, A to D and D to A conversion, Discrete time vs. digital signal processing.	
UNIT II: TRANSFORMED DOMAIN ANALYSIS OF DISCRETE TIME SIGNALS AND SYSTEMS	**10 Hrs.**
Z-transform, ROC and Properties, Poles- Zeros, Inverse z-transform, LTI System Analysis using Z-transform, Discrete Time Fourier Transform (DTFT) and Important Properties, Frequency Response of Discrete Time Systems, Correlation of Discrete Time Signals, Parseval‟s relation, Energy and Power Spectral Density.	
UNIT III: DFT AND FFT ALGORITHMS	**10 Hrs.**
Effect of periodicity and discretization on spectra of a signal, Sampling of DTFT, DFT and IDFT, Important Properties of DFT, Linear and Circular Convolution, Application of DFT in Linear Filtering, Efficient Computation of DFT, Radix2 FFT Algorithms, Geortzel Algorithm.	
UNIT IV: DIGITAL FILTERS: DESIGN AND STRUCTURES	**11 Hrs.**
Ideal Digital Filters, Practical Filters: Stability and Causality, FIR and IIR Filters, Linear Phase and Implications, Filter Design Steps, Design of Linear Phase FIR Filters, Window Method, IIR Filter Design, Pole-Zero Placement Method, Overview of Laplace Transform and Analog Filter Design, Analog Filter Standard Frequency Responses and Design Equations, IIR Filter Coefficients from Analog Filter, Bilinear Transformation, Basic Structures for FIR and IIR Systems implementation, Direct, Transposed Cascade and Parallel form Structures, Effects of Co-efficient Quantization.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember properties of discrete time signals and systems in Time domain, Z-domain and frequency
CO2	:	Understand concepts and complexity of digital signal processing techniques.
CO3	:	Apply concepts to realize discrete time systems and digital signal processing algorithms.
CO4	:	Analyze discrete time systems, algorithms and digital filters.
CO5	:	Evaluate performance of discrete time systems and digital signal processing algorithms.
CO6	:	Design and implement practical DSP applications.

TEXT/REFERENCE BOOKS

1. Proakis, Manolakis, “Digital Signal Processing: Principles, Algorithm & Application”, Pearson Education.
2. Emmanuel Ifeacher,, Barrie W..Jervis, “Digital Signal processing-A Practical Approach”, Pearson Education.
3. Allen V. Oppenheim, Ronald W. Schafer, “Discrete Time Signal Processing”, Prentice Hall.
4. P. Ramesh Babu, “Digital Signal Processing”, Scitech Publications (India) Pvt Ltd

24ECE***P					Digital Signal Processing Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To impart knowledge of flow process of digital signal processing.
- To implement digital signal processing algorithms on software and hardware platforms.
- To evaluate performance of digital signal processing systems and algorithms.
- To analyze the performance of DSP algorithms on DSP processors.

LIST OF EXPERIMENTS	
1	Introduction to simulation tool and environment: Command Window, Variables, Constants, Scalars, Matrices and Vectors, Functions, Comments, Command History, Workspace, Editor, Script file etc.
2	Discrete time signal generation.
3	Discrete time system analysis in Z-domain and frequency domain.
4	Linear Convolution using Overlap-add and overlap-save method.
5	Circular convolution of two given sequence.
6	Moving average filter analysis.
7	FIR filter design.
8	IIR filter design –I.
9	IIR filter design – II and analysis.
10	Implementation of Decimation and Interpolation processes.
11	Introduction to DSP Processor, DFT of a discrete time signal using DSP Processor.
12	Implementation of FFT of a given sequence using DSP processor.
13	Implementation of LP and HP IIR/FIR filter using DSP Processor.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember representation of discrete time signals and systems in Time domain, Z-domain and frequency domain.
CO2	:	Understand the methodologies to utilize digital signal processing techniques practically.
CO3	:	Apply concepts to realize discrete time systems and digital signal processing algorithms practically.
CO4	:	Analyze and report the behaviour of discrete time systems and digital signal processing algorithms.
CO5	:	Evaluate and report performance of discrete time systems and digital signal processing algorithms.
CO6	:	Design and implement practical applications using various tools.

TEXT/REFERENCE BOOKS

1. Proakis, Manolakis, **"**Digital Signal Processing: Principles, Algorithm & Application", Pearson Education.
2. Emmanuel Ifeacher,, Barrie W..Jervis, "Digital Signal processing-A Practical Approach", Pearson Education.
3. Allen V. Oppenheim, Ronald W. Schafer, "Discrete Time Signal Processing", Prentice Hall.
4. P. Ramesh Babu, "Digital Signal Processing", Scitech Publications (India) Pvt Ltd